

Architecting for Demo Day: A Full-Stack Blueprint for Robust Geospatial AI Under Hackathon Constraints

Database & State Management Architecture

The architectural foundation of this geospatial AI platform rests upon a multi-layered database and state management strategy designed for high-performance analytical queries, real-time transactional integrity, and low-latency caching. This design directly counters the primary technical challenge of handling near-real-time thermal anomalies at scale by employing a pre-computed feature store paradigm. The system utilizes three distinct database technologies—PostgreSQL/PostGIS, DuckDB, and Redis—each chosen for its specific strengths in handling different stages of the data lifecycle, from raw ingestion to analytical querying and real-time inference lookups ^{78 125}. This hybrid approach ensures that each component operates within its domain of excellence, preventing bottlenecks and ensuring the entire system can meet stringent performance targets, such as sub-10ms spatial query latencies and sub-50ms inference times per hotspot ^{74 83}.

The core of the system is the **H3 Hexagonal Feature Store**, which serves as the universal key for all geospatial data fusion. This store is primarily housed in DuckDB, leveraging its in-process, columnar architecture optimized for analytical workloads ^{84 296}. DuckDB provides a lightweight yet powerful solution for managing large analytical datasets locally or within a process, making it ideal for the static contextual features derived from OpenStreetMap (OSM) and WRI Global Forest Watch data ¹⁴⁸²⁷⁵. The feature store is populated offline by the ML/Data team, who perform computationally intensive tasks like raster zonal statistics and vector distance transforms once a week or month ^{1 5}. By pre-computing these features and storing them in DuckDB, the system eliminates the need for expensive real-time spatial joins during inference, a critical decision for achieving the <50ms latency requirement ¹⁰⁶²⁴⁰. DuckDB's native support for H3 through its community extensions allows for direct SQL-based manipulation of the grid, including functions to convert coordinates to H3 cells (`h3_latlng_to_cell`), find neighboring cells (`h3_grid_disk`), and aggregate data efficiently ^{2 85 112}. The schema for this feature store is designed for fast lookups, with the H3 cell ID as the primary key and a

dense set of pre-computed numerical and categorical features representing land cover, proximity to infrastructure, and other contextual attributes ³.

For raw data ingestion, PostgreSQL with the PostGIS extension serves as the authoritative source of truth. Its robust transactional capabilities, ACID compliance, and mature ecosystem make it the ideal choice for ingesting and storing the high-frequency stream of NASA FIRMS point data ²⁹⁰²⁹¹. The raw FIRMS data, along with any other transient or frequently updated vector data, is loaded into a dedicated schema within PostgreSQL. This layer handles the variability and potential schema drift of incoming public APIs, providing a stable interface for downstream processes ^{6 7}. To ensure performance, a rigorous indexing strategy is paramount. For spatial filtering, GiST (Generalized Search Tree) indexes on geometry columns are essential, as they use R-tree algorithms to group geometries based on their bounding boxes, enabling rapid filtering of candidate rows ^{267 316317}. Given the time-series nature of the FIRMS data, BRIN (Block Range Index) indexes on the timestamp columns are also employed. BRIN indexes store summary information about value ranges for blocks of table pages, offering significant space savings over traditional B-tree indexes and performing well for large, ordered datasets like time-series data ²⁶⁶³¹⁵. Together, a GiST index for spatial queries and a BRIN index for temporal queries provide a powerful combination for fast filtering on the raw data table. Further optimization for bulk inserts can be achieved by temporarily disabling triggers and constraints, batching transactions, and tuning PostgreSQL's Write-Ahead Log (WAL) settings to reduce I/O overhead during data loading ²¹⁷³¹⁴³⁷³.

Finally, Redis is integrated into the architecture as a high-speed, in-memory caching layer to serve the most frequently accessed portions of the H3 feature store ¹⁶⁵. While DuckDB is highly performant, for the hottest H3 keys (i.e., those corresponding to areas with the highest density of recent fire events), a Redis lookup can offer even lower latency, often approaching sub-millisecond response times ²⁵⁰. This tiered caching strategy ensures that the most common context vectors are available with minimal delay, further solidifying the system's ability to meet the strict real-time inference requirements. The integration between PostgreSQL and DuckDB can be seamless; for instance, data can be extracted from PostgreSQL tables containing OSM or raster-derived features and loaded into DuckDB for the final feature store assembly ^{5 9}. The following table details the specific roles and technologies for each state management component.

Component	Technology	Primary Role	Key Features & Justification
Raw Data Storage	PostgreSQL with PostGIS	Authoritative storage for ingested NASA FIRMS data and other vector sources.	Transactional integrity, ACID compliance, robustness for high-frequency writes, mature geospatial extensions ²⁹⁰ ²⁹¹ .
Analytical Feature Store	DuckDB	High-performance, in-process storage for the pre-computed H3 hexagonal feature store.	Columnar, vectorized execution for analytical queries ⁸⁴ ; native H3 and spatial extensions for efficient feature engineering ⁸⁵ ²¹¹ ; lightweight and embeddable ²⁹⁶ .
Real-Time Caching Layer	Redis	Ultra-low-latency caching of the most frequently accessed H3 feature vectors.	In-memory key-value store with sub-millisecond read/write performance, reducing pressure on DuckDB for hot data ¹⁶⁵ ²⁵⁰ .

This carefully architected data layer forms the bedrock of the system, transforming complex, real-time geospatial analysis into a series of simple, lightning-fast lookups. It directly addresses the challenges of ingestion choke points and high-latency inference by offloading all heavy computation to an offline batch process, thereby creating a resilient and scalable foundation for the rest of the full-stack application.

Backend Concurrency and Asynchronous Ingestion Pipeline

To withstand the immense load of a sudden influx of 20,000 new FIRMS records and prevent the "Ingestion Choke" failure mode, the backend must be architected for non-blocking, asynchronous processing. The proposed solution leverages a modern Python stack centered around FastAPI, Celery, and Redis to create a resilient ingestion pipeline that keeps the primary inference API responsive under all conditions ³¹ ¹⁸³. This architecture decouples the long-running task of bulk data insertion from the short-lived HTTP request-response cycle, a fundamental principle for building scalable and robust services ³² ³⁵⁷. When NASA's FIRMS service delivers a massive CSV file, the FastAPI endpoint acts as a message gateway. It first validates the incoming data format and schema, then immediately acknowledges receipt and enqueues the task to be processed asynchronously by a pool of Celery workers ¹⁹⁵³⁶⁹. The communication between FastAPI and the workers is managed via Redis, which serves as both the message broker and, optionally, the result backend ²⁵¹³⁸⁶. This separation ensures that the main FastAPI process, typically running on an ASGI server like Uvicorn, is never tied up waiting for a potentially slow database write operation, thus avoiding timeouts and maintaining a healthy request queue for legitimate inference calls ¹⁹⁸.

The ingestion process itself involves several best practices to maximize throughput and minimize database strain. Upon receiving the task from the FastAPI gateway, a Celery worker initiates a single, long-running transaction to perform the bulk insert into the PostgreSQL/PostGIS database ^{70 354}. Before the bulk load begins, the script will disable any non-essential constraints and triggers on the target table to eliminate overhead ^{217 373}. For optimal performance, it is recommended to use batch inserts, for example, committing every 10,000 records, rather than one massive transaction, which helps manage memory usage and reduces rollback time if an error occurs ⁷⁰. After the data has been successfully loaded, the script re-enables the previously disabled constraints and triggers ²¹⁷. Finally, to further accelerate the process, the `COPY` command in PostgreSQL is preferred over individual `INSERT` statements, as it is specifically optimized for bulk data loading from files or streams ²¹⁹. This entire sequence—disabling constraints, bulk loading with `COPY`, and re-enabling constraints—is wrapped in a transactional block, ensuring atomicity; if any part fails, the entire operation can be rolled back cleanly ²⁹³. The use of explicit transactions for bulk operations is a proven technique for significantly improving write performance in PostgreSQL ²¹⁵²¹⁸.

A critical component of this defensive architecture is the schema validation layer. Public APIs, especially from government agencies like NASA, can occasionally introduce schema changes or deliver malformed data ²⁷⁶. To protect the ingestion pipeline from such out-of-band changes, the initial FastAPI endpoint should employ Pydantic models to strictly validate the incoming JSON payload structure ³⁴⁵. Any record that fails this validation should not be processed but instead routed to a dead-letter queue (DLQ), which could be another Redis list or a dedicated table in PostgreSQL ⁶. Logging these schema violations provides invaluable debugging information and prevents corrupt data from polluting the main dataset, which could otherwise lead to unpredictable behavior in the machine learning model or downstream analytics ⁷. This pattern of validating inputs and isolating bad data is a cornerstone of resilient data engineering ²⁶³. The entire process is orchestrated by Celery, which manages task distribution, retries, and monitoring, allowing the team to focus on the core logic of the ETL process rather than the complexities of distributed task management ¹⁹⁶²⁰⁰.

The API contract between the frontend/UI and the ML backend is meticulously defined to ensure loose coupling and independent development cycles. As established in the architectural vision, the backend's role is not to pass raw coordinates but to provide a fully assembled feature payload ³⁶⁴. The ingestion pipeline is responsible for enriching the raw FIRMS data with the H3 index and other necessary metadata before it is stored. The `/api/v1/classify` endpoint on the FastAPI server expects a JSON object

structured as specified in the PRD, containing both the FIRMS-specific features and the pre-fetched spatial context from the H3 feature store ³⁶⁴. This rigid contract, formally documented in an OpenAPI specification, acts as a buffer between teams. The UI team can evolve their frontend logic without needing to understand the internal mechanics of the H3 conversion or the feature store lookup, and the ML team can modify the feature engineering logic in the backend without impacting the frontend's ability to submit classification requests, provided the JSON schema remains consistent ³⁶⁴. This separation of concerns is crucial for agility within a tight 3-week sprint. The following table outlines the key components of this asynchronous architecture.

Component	Technology	Role in Ingestion Pipeline	Justification
Web Framework	FastAPI	Receives incoming FIRMS data, performs initial validation, and enqueues background tasks.	High performance, automatic OpenAPI generation, excellent async support, Pydantic integration for data validation ^{31 345} .
Message Broker	Redis	Acts as the communication channel between the FastAPI app and Celery workers.	Lightweight, in-memory data structure store; supports pub/sub patterns for task distribution ²⁵¹³⁸⁶ .
Task Queue	Celery	Manages the execution of long-running background jobs, including bulk inserts and data enrichment.	Distributed task queue system; handles job scheduling, retries, and worker management for Python applications ¹²³³⁶⁹ .
Database	PostgreSQL/PostGIS	Stores the final, enriched dataset of FIRMS hotspots.	Reliable, transactional storage with powerful PostGIS extensions for spatial data handling ²⁹⁰²⁹¹ .
Data Validation	Pydantic	Validates the structure and schema of incoming data from the FIRMS API.	Provides data parsing and validation based on Python type hints; integrates seamlessly with FastAPI ³⁴⁵ .

By implementing this decoupled, asynchronous architecture, the system becomes inherently resilient to the "Ingestion Choke." It gracefully handles high-volume data bursts, maintains low-latency responses for its primary function (inference), and builds in safeguards against common data quality issues, forming a robust backbone for the entire application.

Frontend WebGL Rendering and Performance Optimization

The "Browser Crash" failure mode, stemming from the DOM overload of attempting to render tens of thousands of geospatial features, is a common pitfall in hackathon projects. The proposed solution is to completely bypass traditional browser rendering methods and adopt a GPU-accelerated visualization toolkit like Deck.gl ^{53 145}. Standard

mapping libraries such as Leaflet.js are fundamentally limited because they rely on manipulating the Document Object Model (DOM) to display markers and shapes. Each element added to the DOM incurs a performance cost, and rendering tens or hundreds of thousands of elements quickly exhausts the available memory and CPU resources of a typical laptop, leading to unresponsiveness or a complete crash ^{44 145}. Deck.gl, on the other hand, is a WebGL-powered framework designed explicitly for rendering large-scale geospatial datasets ¹⁷⁷¹⁹³. It renders visualizations directly onto an HTML5 `<canvas>` element using the GPU, which is massively parallel and excels at the repetitive tasks of drawing millions of similar geometric primitives, such as hexagons or points ^{53 177}. This offloading of work from the CPU to the GPU is the single most important factor in achieving the required 60 frames-per-second (FPS) performance while visualizing 100,000+ H3 hexagons and their associated SHAP pop-ups ³⁰⁵.

However, simply using Deck.gl is not enough to guarantee performance on standard hardware. The system must implement sophisticated client-side optimization strategies to manage the data sent to the browser. The two most critical techniques are viewport-culling and dynamic clustering ¹. Viewport-culling ensures that the frontend only requests data for the currently visible portion of the map. Instead of fetching the entire dataset from the backend, the frontend code listens for map movement and zoom events. When the view changes, it calculates the bounding box of the new viewport and makes an API call to the backend, passing this bounding box as a parameter. The backend then executes a spatial query against the PostgreSQL database to return only the H3 hexagon features whose geometries intersect with the requested area ¹. This dramatically reduces the volume of network traffic and the amount of data the browser needs to parse and process, preventing timeouts and memory overflows. The backend API should be designed to accept spatial parameters (e.g., Well-Known Text or GeoJSON of the bounding box) and leverage the GiST indexes created earlier to execute this query efficiently.

Dynamic clustering complements viewport-culling by adapting the level of detail based on the user's zoom level. At a high-level view, such as looking at the entire country of India, rendering individual H3 Resolution 8 hexagons would be visually overwhelming and computationally wasteful. Instead, the frontend should request aggregated cluster data, perhaps from coarser H3 resolutions (e.g., Resolution 4 or 5), which would show larger polygons representing groups of smaller hexagons, annotated with counts of fire events ¹. As the user zooms in progressively, the frontend can trigger new API calls with finer-grained bounding boxes and request data at higher H3 resolutions (e.g., Resolution 8). This hierarchical approach, enabled by the intrinsic properties of the H3 grid system, ensures that the number of rendered objects is always appropriate for the screen real

estate and the user's intent ³ ⁹⁰. Libraries like `h3-py` can assist in compacting adjacent hexagons of the same value into larger parent cells, further simplifying the visualization at low zoom levels ¹. The `Deck.gl` library itself provides powerful layers for this purpose, such as the `H3HexagonLayer` for rendering the hexagons and a custom `ScatterplotLayer` or `TextLayer` for displaying the count labels on top of clusters ¹ ⁴⁵.

Furthermore, the system must be prepared for unexpected hardware limitations. A judge's laptop might have limited VRAM, causing a WebGL context loss error, particularly after rapid zooming or panning ²⁰ ²¹. The frontend code should implement a graceful degradation strategy. It can periodically check device memory and performance metrics. If a WebGL context loss event is detected (`webglcontextlost`), the application should first attempt to recover by recreating the `Deck.gl` instance, as the browser may be able to restore the context after a brief pause ¹⁸ ²². If recovery is not possible or if the device is identified as having very limited resources upfront, the frontend can fall back to a simpler rendering mode, such as displaying clustered Mapbox vector tiles instead of the rich `Deck.gl` visualizations ⁴⁴. This proactive handling of edge cases demonstrates a deep understanding of frontend engineering and ensures the demo remains stable and impressive regardless of the judge's hardware. The configuration of the `Deck.gl` application itself is straightforward, involving the creation of a `Deck` instance and adding the relevant layers, such as the `H3HexagonLayer`, which takes an array of H3 cell data and maps properties like color and opacity to the data fields ¹ ²³. This comprehensive strategy, combining a powerful rendering engine with intelligent data management and graceful degradation, effectively neutralizes the "Browser Crash" threat and enables a smooth, interactive experience even with massive datasets.

DevOps, Containerization, and Disaster Recovery

The "Geospatial Dependency Hell" is a notorious challenge where installing Python geospatial libraries like `geopandas`, `rasterio`, and `shapely` fails due to missing or incompatible C++ dependencies such as GDAL, GEOS, and Proj ⁴⁰ ⁴². The definitive solution is to leverage official, pre-built Docker images from the Open Source Geospatial Foundation (OSGeo) ¹⁷⁹. Instead of starting with a generic Linux image (like `ubuntu` or `python:3.9-slim`) and attempting to compile these complex libraries from source, the Dockerfile should begin with a targeted base image like `osgeo/gdal:ubuntu-small` or `osgeo/gdal:alpine-small` ³⁴ ³⁷. These images are meticulously maintained by

the OSGeo community and contain GDAL and other essential libraries compiled and linked correctly, providing a stable and reproducible environment ¹¹⁷. The subsequent steps in the Dockerfile can then install Python packages like `rasterio` and `geopandas`, which will link against the already-present system-level GDAL installation, completely bypassing the compilation errors that plague naive approaches ^{40 181}. This method results in leaner, more secure containers compared to bloated images that bundle multiple compilers and build tools. The Dockerfile should also include best practices like using multi-stage builds to separate the build environment from the runtime environment and specifying non-root users to enhance security ¹¹⁹¹²⁰. By adopting this strategy, the team can containerize the entire application stack—including the ML model, the FastAPI backend, and any offline data processing scripts—ensuring consistency across development, staging, and production environments and drastically reducing deployment-related friction ³³⁹.

The Continuous Integration and Continuous Deployment (CI/CD) pipeline, orchestrated with GitHub Actions, plays a critical role in maintaining the system's quality and reliability. The pipeline should be triggered on every pull request and push to the main branch. Its workflow should consist of several key stages. First, a linting and formatting stage to enforce code style consistency. Second, a testing stage that runs a comprehensive suite of unit and integration tests. Crucially, this must include spatial unit tests to validate the correctness of the feature engineering logic. For example, tests should verify that H3 cells are generated at the correct resolution, that zonal statistics from rasters are computed accurately, and that spatial joins produce expected results ⁶³. Tools like `pytest` can be used to run these tests inside a container identical to the production environment, catching environment-specific bugs early. Third, the pipeline should build the Docker image and push it to a container registry. Finally, it should deploy the new version to a staging environment for manual verification. This automated, repeatable process ensures that every change is validated against a production-like setup before it ever reaches the live demo, embodying professional software delivery practices ¹²¹.

The "Disaster Recovery Checklist" is the ultimate expression of pragmatic engineering, preparing the team for the inevitable failures that occur during a high-pressure live demo. The five critical failure scenarios, beyond the initial four failure modes, are:

1. **NASA FIRMS API Blackout:** During the demo, the NASA FIRMS API might become unavailable or rate-limit the IP address ^{57 223}. The fix is to have a local Python script ready that can read from a cached, archived copy of the FIRMS CSV data and feed it into the Redis message queue, simulating a continuous data stream. This requires pre-downloading a historical dataset and scripting the ingestion logic.

2. **The "Zoom Out" OOM Crash:** A judge might ask to "zoom out to see all of India," triggering the backend to query and serialize a massive dataset, likely exceeding API timeout limits ¹¹⁴. The fix is to rigorously enforce viewport-culling at the API gateway. The backend should be coded to reject requests for bounding boxes that are too large and return a small, aggregated overview instead, proving the system is designed for performance at all scales.
3. **SHAP Latency Spike:** Calculating SHAP explanations for the entire visible map view can be computationally expensive and freeze the frontend or backend CPU. The fix is to change the UX flow: SHAP values should only be calculated on-demand. When a user clicks on a specific H3 hexagon, then and only then does the frontend trigger a secondary API call to fetch and display the detailed SHAP explanation for that single hotspot. This avoids a massive synchronous computation.
4. **WebGL Context Loss:** The judge's laptop may run out of VRAM, especially during rapid interactions, causing the browser tab to crash ^{20 21}. The front-end code should listen for the `webglcontextlost` event. On detection, it should attempt to recreate the `Deck.gl` instance. If the problem persists, it should gracefully degrade by switching to a less graphically intensive rendering mode, such as displaying clustered Mapbox vector tiles, ensuring the application remains usable ^{18 22}.
5. **Schema Drift Trap:** NASA might change a column name or data type in the FIRMS CSV feed, breaking the ingestion pipeline ²⁷⁶. The fix is the defensive programming measure of using Pydantic for data validation at the ingestion gateway. Any malformed or schema-incompatible data is caught and logged, routed to a dead-letter queue, and does not disrupt the processing of valid data, preventing a catastrophic failure ^{6 7}.

Having scripts and clear procedures for these five scenarios written and rehearsed before the final deadline is a strategic advantage. It transforms the team from a reactive state to a proactive one, demonstrating calm, professional problem-solving skills under pressure, which is a hallmark of winning solutions.

NTRO Compliance and Adversarial Resilience

Addressing the "NTRO Audit Gap" requires moving beyond simple logging to establish a verifiable chain of custody for every decision made by the AI system. For the 3-week timeframe, the most pragmatic and effective approach is to implement an immutable, append-only audit trail within the PostgreSQL database, augmented with strict Role-

Based Access Control (RBAC) 337388. An audit log table is created in the database schema, designed to capture every override action performed by a human analyst. This table must contain a rich set of metadata for each entry, including a unique `event_id`, a `timestamp` with microsecond precision, the `analyst_id` (enforced via RBAC), the original AI prediction, the `action` taken (e.g., "Override: Wildfire -> Industrial"), and a free-text `justification` field where the analyst can enter their reasoning 333389. The immutability of this log is enforced at the database layer itself: the application's database user account is granted `SELECT` privileges, but `UPDATE` and `DELETE` privileges are revoked entirely 14. This means that once a log entry is inserted, it cannot be altered or deleted by any application logic, providing a tamper-evident record that satisfies core tenets of compliance standards like NIST SP 800-171 and GxP 333391. While full cryptographic signing is an advanced feature, this database-enforced immutability provides a strong, production-grade baseline for auditability 389.

The system's resilience extends to its machine learning pipeline, particularly in preventing the "Leakage Trap," where features engineered from spatial data inadvertently leak information from the future or neighboring areas into the training set, leading to overly optimistic and misleading performance metrics 69. To combat this, the project mandates the use of Spatial Cross-Validation (SCV) instead of standard K-Fold CV 71. SCV is a specialized validation technique that respects the spatial autocorrelation inherent in geographical data. Instead of randomly shuffling and splitting data, SCV ensures that the geographic regions used for the training folds are separated from the regions used for the validation folds by a minimum distance (e.g., 50km) 71. This prevents the model from "cheating" by learning patterns from a location that is spatially proximate to the test location, thereby providing a more realistic estimate of its generalization performance on unseen geographical areas. This rigorous validation strategy is a critical step in building a trustworthy and defensible model for an intelligence agency audience.

Finally, the architecture is designed with "NTRO-portability" in mind, acknowledging the reality of a greenfield cloud-native development within a constrained timeline 332344. The team will build a modern, containerized system using standard protocols like REST APIs and structured JSON logs. However, the pitch to the judges is that this architecture is not locked into a public cloud. The entire system is packaged in Docker containers, and all inter-service communication is handled via well-defined, stateless APIs. This modular, containerized design means that deploying the system into NTRO's air-gapped, on-premises Kubernetes clusters would require no code refactoring—only replacing the public cloud hosting infrastructure with private cloud equivalents 344. The inclusion of standard authentication stubs (e.g., OIDC/OAuth2), strict RBAC enforcement in the application code, and centralized, structured logging are deliberate choices that align

with enterprise-grade security and operational practices expected by organizations like NTRO ³⁸⁸³⁹⁰. This approach demonstrates a mature understanding of defense tech systems: building a flexible, portable, and compliant architecture that can adapt to different deployment environments without compromising its core functionality or security posture.

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